

Food Waste Donation Management System

Connecting Surplus Food with Communities in Need — Bangladesh

Submitted by Team

CODE FOR GOOD

Java · MySQL · JDBC · Machine Learning
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Team Members

This project has been designed and developed by the following team members under the team name **Code For Good**, as part of the Food Waste Donation Management System initiative.

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1. Introduction

Every year, a staggering amount of edible food is thrown away by restaurants, hotels, grocery stores, caterers, and households across Bangladesh, even as millions of people struggle with food insecurity. According to the Food and Agriculture Organization (FAO) and the UNEP Food Waste Index, roughly one-third of all food produced globally is lost or wasted, while hunger and malnutrition remain pressing issues in many communities, including low-income neighborhoods in Bangladesh. This mismatch between surplus food and unmet need is not a scarcity problem — it is a distribution and coordination problem. The **Food Waste Donation Management System** is a digital platform designed to bridge this gap by connecting surplus-food providers (restaurants, event organizers, grocery stores, and individuals) with NGOs, shelters, orphanages, and underprivileged communities that can make use of that food before it goes to waste.

1.1 Bangladesh Food Waste Landscape (Latest Statistics, 2026)

Recent government and international reports paint a clear picture of the scale of the problem in Bangladesh. In June 2026, the State Minister for Food told Parliament that Bangladesh wastes around 3.5 million tonnes of food annually, citing the UNEP Food Waste Index Report. A World Bank study presented in Dhaka found that the country wastes nearly 34% of its available food each year — equivalent to roughly 4% of national GDP — with more than 27% of cultivated farmland (over 34,000 sq. km) growing food that is ultimately never eaten. Food loss and waste also account for about 13% of Bangladesh's total greenhouse gas emissions, close to 29 million tonnes of CO2 equivalent annually. A Bangladeshi household wastes an estimated 82 kg of food per person every year — higher than the United States, the Netherlands, or Japan. A 2026 consumer-behavior survey of 1,216 respondents further found that higher-income households waste about 32% more food than lower-income groups, while 47% of respondents discard food based on appearance alone and 39% misinterpret expiry labels, highlighting both a logistics gap and an awareness gap that a donation-matching platform can help address.

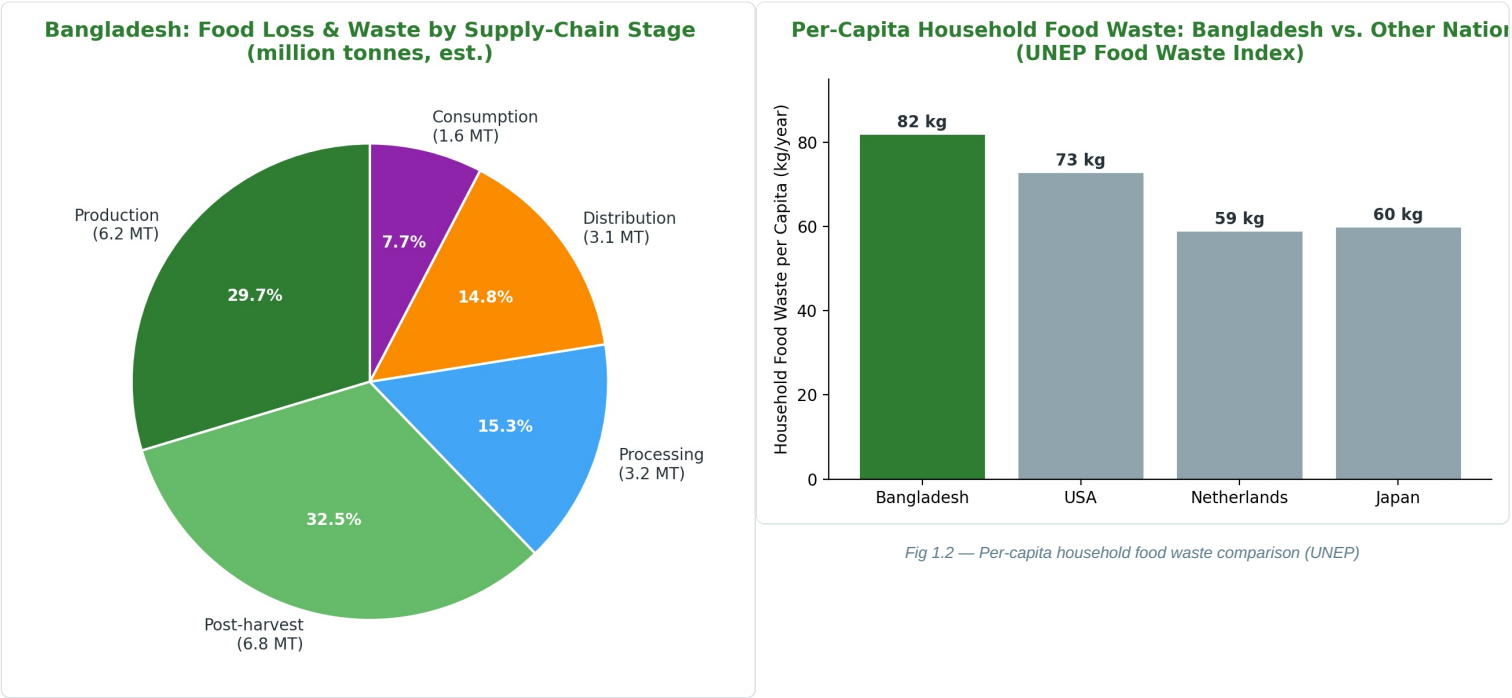


Fig 1.1 — Estimated food loss/waste by supply-chain stage in Bangladesh

Fig 1.2 — Per-capita household food waste comparison (UNEP)

Bangladesh: Estimated Food Loss Rate by Commodity

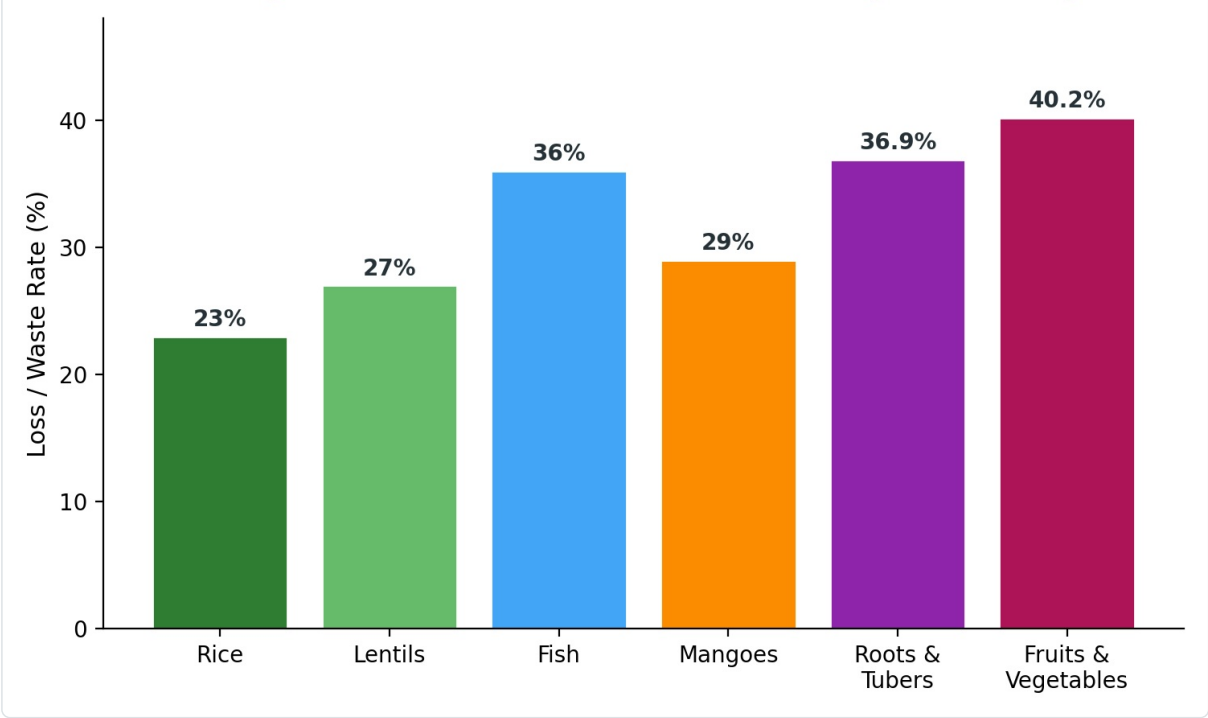


Fig 1.3 — Estimated loss rate by commodity type in Bangladesh

2. Problem Statement

Large quantities of safe, edible food are discarded daily by restaurants, caterers, wedding venues, and grocery stores in Bangladesh simply because there is no efficient, trustworthy, and immediate channel to redirect that surplus to people who need it. NGOs and shelters, on the other hand, often struggle to locate reliable sources of surplus food in real time, relying instead on word-of-mouth or occasional donation drives. This lack of a centralized, real-time coordination system results in:

- Increased food waste and the associated environmental burden, including landfill methane emissions;
- Continued hunger among vulnerable populations despite food surplus existing nearby;
- Missed opportunities for businesses and individuals to contribute to social good in a simple, trackable way.

The core problem, therefore, is the **absence of a fast, transparent, and accessible system** to match surplus food with the people and organizations who need it most, before that food spoils and its value is lost entirely.

3. Objectives

3.1 Reducing Food Waste

Enable restaurants, event organizers, and households to list surplus food in real time so it can be collected and redistributed before it spoils, rather than being thrown away.

3.2 Supporting Food-Insecure Communities

Give NGOs, shelters, and community volunteers a reliable, up-to-date feed of available food donations nearby, along with pickup logistics, so they can consistently serve people in need.

3.3 Encouraging Corporate & Individual Social Responsibility

Provide businesses and individuals with an easy, low-friction way to donate surplus food and track their contribution and impact over time, encouraging a culture of giving rather than discarding.

3.4 Intelligent Matching

Use a data-driven recommendation engine to automatically suggest the most suitable NGO for each donation, minimizing pickup delay and maximizing the chance that food is collected before it spoils.

4. Preliminary Solution

Before finalizing the detailed system architecture, the team outlined the following preliminary solution concept — the initial set of ideas identifying what the platform needed to include at a high level:

- A **donor-facing module** where restaurants, grocery stores, event organizers, and individuals can post surplus food listings (type, quantity, condition, and pickup window).
- A **recipient-facing module** for verified NGOs, shelters, and volunteers to browse and claim nearby donations in real time.
- A **location-based matching and notification system** that alerts nearby verified recipients as soon as a new donation is posted.
- A **verification and rating system** to ensure food safety, donor accountability, and recipient trustworthiness.
- An **impact-tracking dashboard** showing donors and the platform how much food (by weight/meals) has been saved and how many people have been served.

This preliminary concept was later expanded into the detailed layered architecture, functional/non-functional requirements, UML diagrams, database design, and the ML-based NGO recommendation engine described in the following sections.

5. Literature Review

Food rescue and donation-matching platforms have gained global attention as a practical response to the twin problems of food waste and food insecurity. Applications such as *Too Good To Go* (Europe) and *Food Rescue US* (United States) demonstrate that a simple listing-and-claim model, supported by geolocation, can meaningfully reduce food waste when donors and recipients are digitally connected. Academic and institutional research — including the FAO's global food-loss estimates and the UNEP Food Waste Index Report 2024 — consistently finds that the majority of food loss in developing countries like Bangladesh occurs due to weak cold-chain infrastructure, poor coordination between surplus producers and needy recipients, and limited real-time visibility of available food. A 2026 Bangladesh-focused consumer survey found that behavioral factors (appearance bias, expiry-label misunderstanding) compound the logistics problem, suggesting that any technical solution should combine efficient matching with transparency and simple verification.

Existing donation efforts in Bangladesh remain largely informal, driven by phone calls, social-media posts, or occasional charity drives, with no centralized, verifiable, or trackable system. This creates three recurring gaps that the literature consistently identifies: (1) lack of real-time discovery of surplus food, (2) lack of trust/verification between anonymous donors and recipients, and (3) lack of measurable impact tracking to sustain donor motivation. The proposed Food Waste Donation Management System directly targets these three gaps using a web/desktop Java application backed by a relational database, with an added machine-learning layer for intelligent NGO matching — an enhancement not commonly found in the simpler rule-based matching used by many existing regional food-rescue tools.

6. Proposed Solution & System Architecture

The proposed system is built around five core modules:

- **Donor Module** — restaurants, grocery stores, event organizers, and individuals can post surplus food listings (type, quantity, condition, and pickup window).
- **Recipient (NGO) Module** — verified NGOs, shelters, and volunteers can browse and claim nearby donations in real time.
- **Location-based Matching & Notification** — alerts nearby verified recipients as soon as a new donation is posted, ranked using the ML recommendation engine.
- **Verification & Rating System** — ensures food safety, donor accountability, and recipient trustworthiness through a review mechanism.
- **Impact-Tracking Dashboard** — shows donors and administrators how much food (by weight/meals) has been saved and how many people have been served.

Food Waste Donation Management System — Layered Architecture

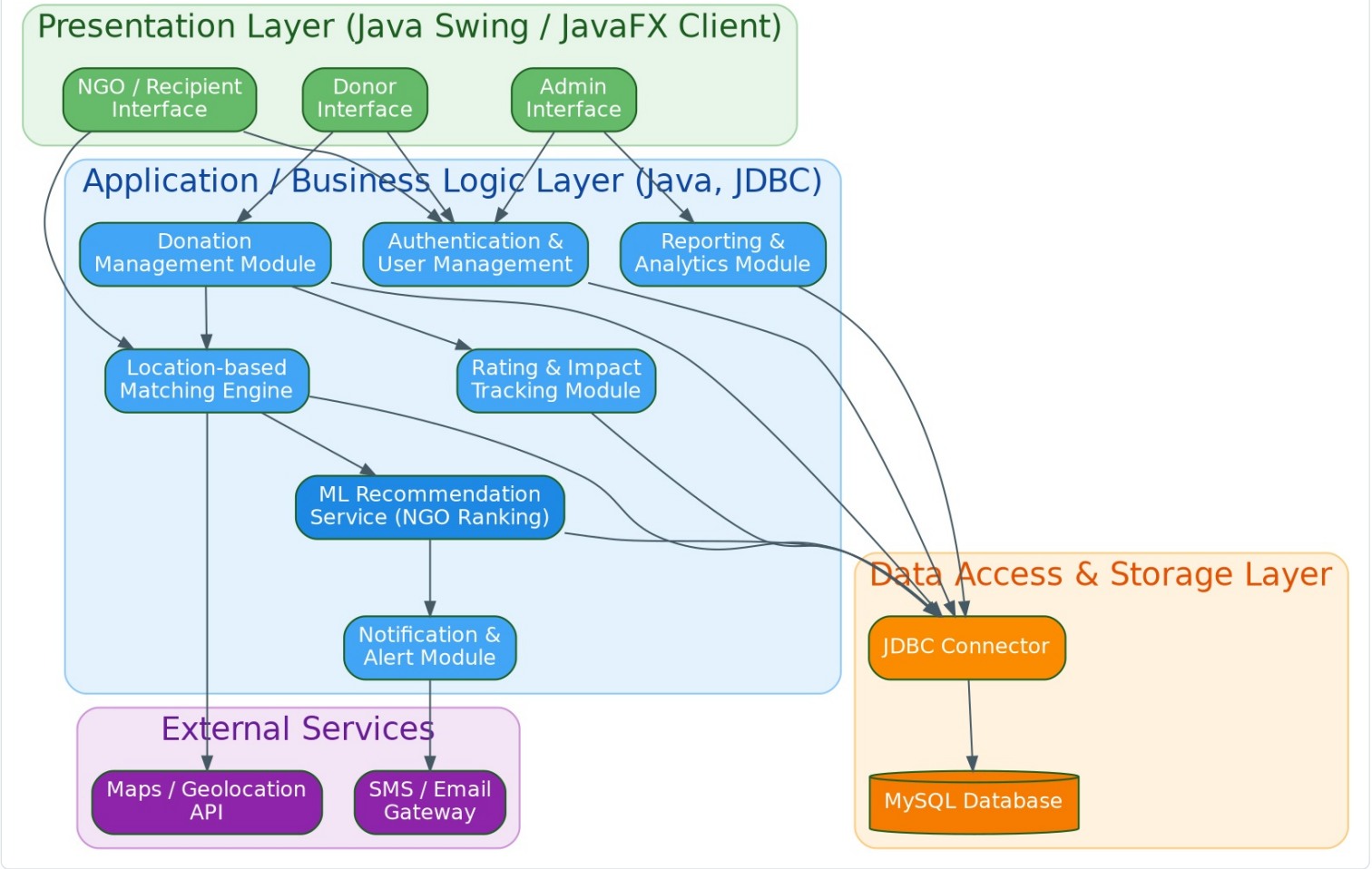


Fig 6.1 — Layered system architecture: Presentation, Application/Business Logic, Data Access, and External Services

The architecture follows a classic layered (n-tier) design suited to a Java desktop/web application: a **presentation layer** built with Java Swing/JavaFX for Donor, NGO, and Admin interfaces; an **application layer** containing authentication, donation management, the matching engine, the ML recommendation service, notifications, ratings, and reporting; a **data access layer** using JDBC to communicate with a MySQL database; and integration points with **external services** such as a maps/geolocation API and an SMS/email gateway for notifications.

7. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in as a Donor, NGO/Recipient, or Admin.
FR-02	Donors shall be able to create, edit, and cancel food donation listings (type, quantity, condition, expiry, pickup window, location).
FR-03	NGOs shall be able to browse, filter, and search available donations by location, food type, and expiry time.
FR-04	The system shall automatically rank and recommend the most suitable NGO(s) for each donation using the ML recommendation engine.
FR-05	The system shall send real-time notifications to relevant NGOs when a new matching donation is posted.
FR-06	NGOs shall be able to claim a donation and generate a pickup request, which the donor can confirm.
FR-07	The system shall allow both donors and recipients to rate each other after a completed pickup.
FR-08	The system shall maintain a donation history and an impact dashboard (meals saved, weight donated, CO2 avoided).
FR-09	Admins shall be able to verify/approve new NGO accounts and manage all users.
FR-10	Admins shall be able to generate and export reports (donation volume, active NGOs, regional statistics).

8. Non-Functional Requirements

Category	Requirement
Performance	Donation listings and search results shall load within 2 seconds under normal network conditions.
Scalability	The database schema and application layer shall support growth to at least 50,000 users and 100,000 donation records without redesign.
Security	Passwords shall be stored using secure hashing; role-based access control shall restrict Admin functions to verified Admin accounts.
Reliability	The system shall maintain 99% uptime during operating hours and perform regular automated database backups.
Usability	The interface shall be simple enough for first-time NGO volunteers and small business owners to use without training.
Maintainability	The system shall follow a modular, layered architecture (Java + JDBC + MySQL) to simplify future updates, including mobile and ML enhancements.
Portability	The Java-based client shall run on any major OS (Windows/Linux/macOS) with a compatible JVM.

9. Use Case Diagram (UML)

The use case diagram below identifies three primary actors — **Donor**, **NGO/Recipient**, and **Admin** — along with a supporting system actor, the **ML Recommendation Engine**, which is invoked whenever a donation is posted to identify the best-matching NGO(s).

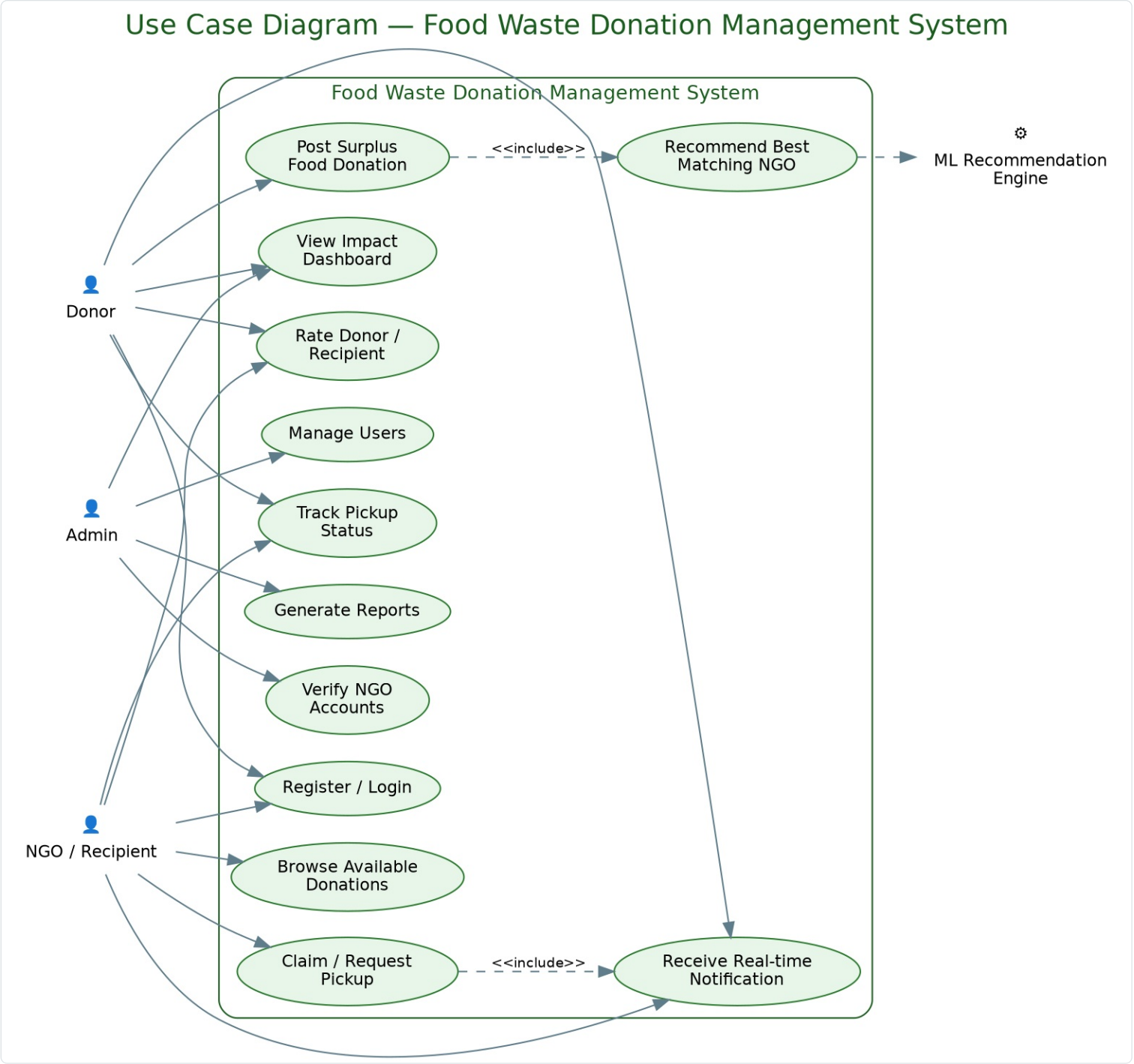
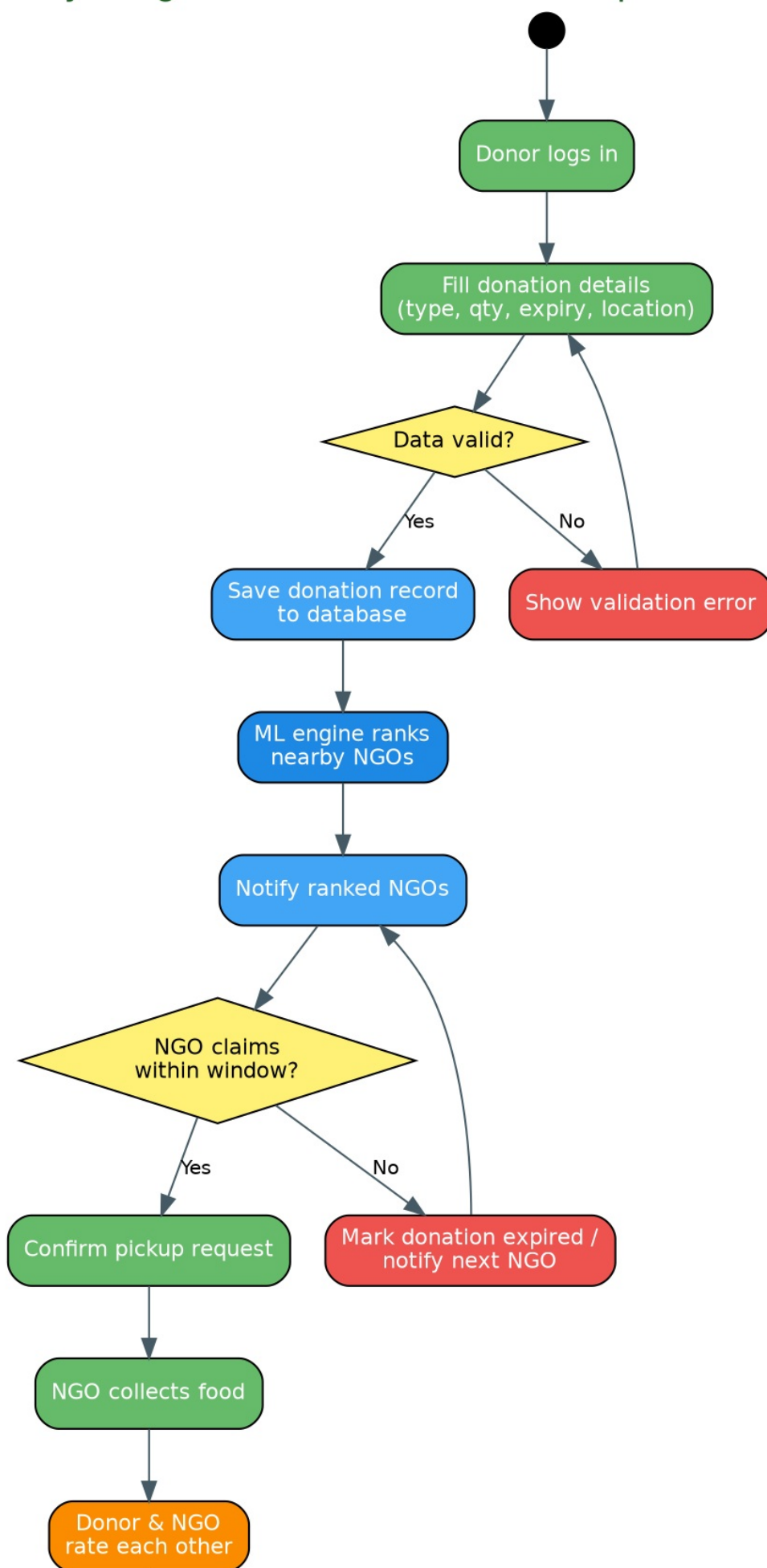


Fig 9.1 — UML Use Case Diagram for the Food Waste Donation Management System

10. Activity Diagram

The activity diagram illustrates the end-to-end workflow from a donor posting a donation through to pickup completion and impact-dashboard update, including the validation and ML-ranking decision points.

Activity Diagram — Donation to Pickup Workflow



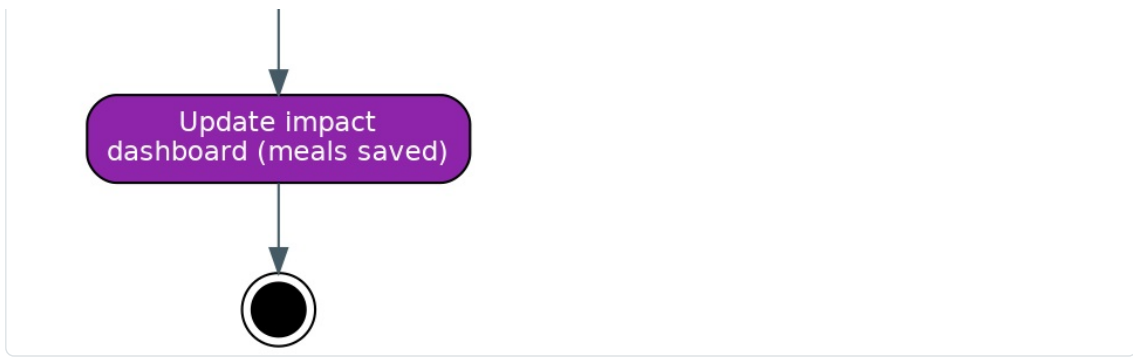


Fig 10.1 — Activity Diagram: Donation-to-Pickup Workflow

11. Database Design (ER Diagram)

The relational schema is designed for MySQL and accessed through JDBC. It consists of six core tables: `USERS` (shared table for Donor/NGO/Admin roles), `DONATION`, `PICKUP_REQUEST`, `RATING`, `NOTIFICATION`, and `IMPACT_LOG`. A single `USERS` table with a `role` column keeps authentication and profile logic centralized while still supporting distinct dashboards per role at the application layer.

Database Design — Entity Relationship Diagram

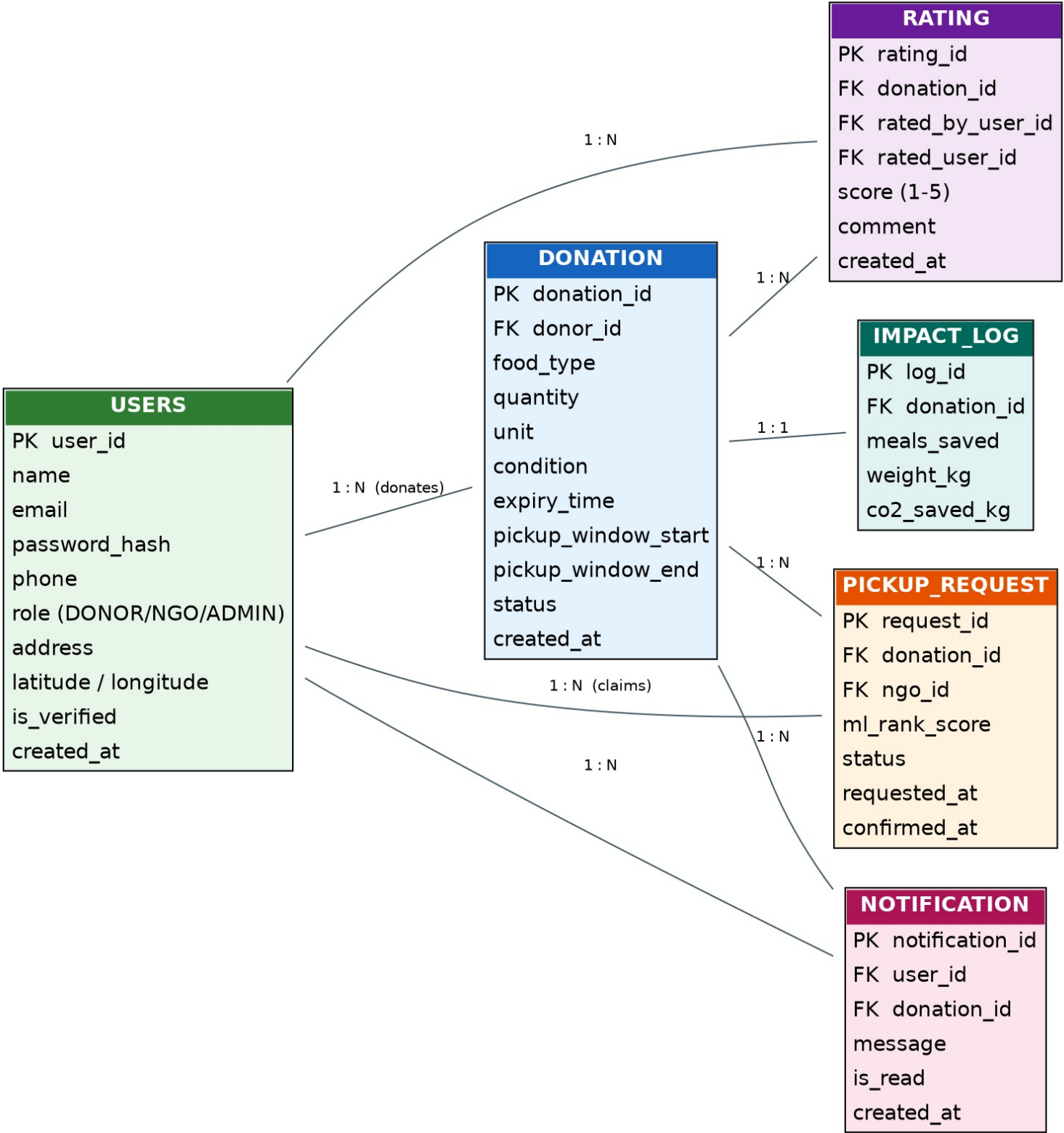


Fig 11.1 — Entity Relationship Diagram

12. UI Mockups

The following wireframes illustrate the intended look and feel of the core screens, designed with a green-and-orange theme reflecting sustainability and food/energy. These act as the low-fidelity Figma/Canva-style mockups guiding front-end development.

Note: These are low-fidelity design mockups created during the design phase, prior to implementation, to plan and guide development — they are static wireframes, not screenshots of a working/built system. No coding, database connection, or functionality has been implemented yet; the actual application will be developed based on these designs in the implementation phase.

FoodShare BD

Reduce Waste. Feed Lives.

Login

Email address

Password

LOG IN

Don't have an account? Register

Register as:

Donor

NGO / Recipient

Volunteer

Fig 12.1 — Login / Registration Screen

Donor Dashboard

Welcome back, Restaurant ABC

1,240

Meals Saved

58

Donations

4.8

Rating

+ POST NEW DONATION

Recent Donations

Cooked Rice & Curry

Qty: 12 kg

Claimed

Bakery Items

Qty: 8 kg

Pending

Fresh Vegetables

Qty: 20 kg

Picked up

Home

Donations

Impact

Profile

Fig 12.2 — Donor Dashboard

Add Donation

Food Type (e.g. Rice, Bread)

Quantity

Unit (kg / plates / packets)

Condition

Expiry / Best-before time

Pickup window (from - to)

Available Donations

Search by area / food type

Nearby

Expiring soon

Large qty

Verified donor

Cooked Rice & Curry

From: Restaurant ABC

Distance: 1.2 km

ML 95 score

CLAIM

Bakery Surplus

From: Sunrise Bakery

Distance: 2.4 km

ML 88 score

CLAIM

Fresh Produce

From: Green Grocers

Distance: 3.1 km

ML 81 score

CLAIM

Pickup location

SUBMIT DONATION

Fig 12.3 — Add Donation Form

Fig 12.4 — Available Donations (NGO View, ML-ranked)

Admin Dashboard — FoodShare BD

4,382

Total Donations

126

Active NGOs

58,140

Meals Saved

31,050

Food Saved (kg)

Pending NGO Verifications

Hope Shelter Trust

APPROVE

City Care Foundation

APPROVE

Basti Relief Society

APPROVE

Reports & Analytics

• Monthly donation volume trend

• Top-performing NGOs by region

• Food waste reduction impact

• User growth (donors vs NGOs)

EXPORT REPORT (PDF)

Fig 12.5 — Admin Dashboard (Verification & Reports)

13. Technology Stack

Component	Technology
Programming Language	Java (JDK 17+)
Database	MySQL 8.x
Database Connectivity	JDBC (Java Database Connectivity)
IDE	NetBeans / IntelliJ IDEA
UI Framework	Java Swing / JavaFX
Version Control	Git & GitHub
ML Layer	Python (scikit-learn) exposed via a lightweight REST API, consumed by the Java application
Diagramming / Design	draw.io (UML & architecture diagrams), Figma/Canva (UI mockups)

14. Machine Learning Integration Plan

The system uses machine learning to solve one specific, well-defined sub-problem: **given a new donation, which registered NGO is most likely to accept and successfully complete the pickup, and how should candidate NGOs be ranked?** This is framed as a **learning-to-rank / recommendation problem**, not a generic classification problem, since the real goal is producing an ordered list of NGO candidates rather than a single yes/no label.

14.1 Recommended Algorithm

Primary model: Random Forest Regressor (scikit-learn `RandomForestRegressor`), used to predict a *match score* (0–100) between a donation and each candidate NGO within a given radius. NGOs are then notified in descending order of predicted score. A Random Forest is recommended over a single Decision Tree because it averages many trees trained on bootstrapped samples, which substantially reduces overfitting and variance — important here because historical pickup data will initially be limited. It also naturally handles a mix of numeric features (distance, quantity) and categorical features (food type, NGO category) without requiring heavy feature engineering, and it can report feature importance, which is useful for explaining why an NGO was recommended.

Baseline/fallback model: A single Decision Tree (or simple weighted-scoring rule) is kept as a fallback for the very first weeks of deployment, before enough historical pickup data exists to train a reliable Random Forest. Once at least a few hundred completed pickup records are available, the system switches to the Random Forest model.

14.2 Input Features

- **Distance** between donor and NGO (km, from geolocation coordinates)
- **Food type match** — whether the NGO's registered preferences/capacity match the donated food category
- **Quantity** of food donated vs. the NGO's typical capacity
- **Time to expiry** — minutes/hours remaining before the food is no longer safe to consume
- **NGO historical pickup rate** — proportion of past claimed donations the NGO actually completed
- **NGO average rating** from donors
- **Time of day / day of week** (captures NGO operating-hour patterns)

14.3 Output & Integration

For each new donation, the trained model scores every eligible NGO within the configured radius and returns a ranked list. The top-ranked NGOs (e.g., top 3–5) are notified first; if none respond within a configurable time window, the next tier is notified automatically. The model runs as a small Python service exposing a single REST endpoint (e.g., `POST /recommend`) that accepts donation + candidate-NGO features as JSON and returns ranked scores; the core Java application calls this endpoint over HTTP and stores the result via JDBC, keeping the ML component decoupled and independently upgradable.

14.4 Training & Evaluation

The model is trained on historical `DONATION` , `PICKUP_REQUEST` , and `RATING` records once sufficient data accumulates (recommended minimum: ~300–500 completed pickups). Data is split 80/20 into training and test sets, with model quality measured using **Mean Absolute Error (MAE)** on the predicted match score and **NDCG (Normalized Discounted Cumulative Gain)** to evaluate ranking quality — i.e., whether the NGOs that actually completed the pickup tended to be ranked near the top. The model should be retrained periodically (e.g., monthly) as new pickup and rating data accumulates.

15. Feasibility Analysis

Type	Assessment
Technical	Java, JDBC, and MySQL are mature, well-documented, and widely taught technologies, making the core system technically low-risk. The ML layer (scikit-learn via a REST API) is a well-established pattern for integrating Python-based models into a Java application.
Economic	The system is built entirely on open-source technologies (Java, MySQL, scikit-learn), minimizing licensing costs. Primary costs are development time, hosting, and NGO onboarding/awareness.
Operational	The interface is designed to be simple for donors and NGO volunteers, most of whom are not technical users; verification and rating mechanisms build the trust needed for sustained adoption.
Schedule	The core system (excluding the ML layer) can be completed within one semester; the ML recommendation feature can be layered in during a second development phase once initial donation data is being collected.

15.1 Estimated Cost

Item	Estimated Cost (BDT)
Development (team effort, tools, testing)	~220,000
Hosting & database (first year)	~60,000
NGO onboarding & awareness campaign	~50,000
Feasibility study (technical/operational/economic)	12,000
Total (tentative budget)	≈ 342,000 (not to exceed 350,000)

15.2 Estimated Timeline

Phase	Duration
Feasibility study	2 weeks
Requirements, UML design & database design	2 weeks
Core module development (Java, JDBC, MySQL)	5 weeks
ML recommendation service integration	2 weeks
Testing, NGO pilot onboarding & rollout	2 weeks
Total estimated duration	≈ 3 months

16. Project Scope

The system manages food donations, NGO requests, pickup tracking, administration, and reporting, with the following boundaries:

- **In scope:** donor and NGO registration/verification, donation listing and browsing, ML-based NGO matching and notification, pickup request/confirmation workflow, rating system, and an impact-tracking dashboard with admin reporting.
- **Out of scope (current phase):** payment processing (the platform is donation-only, not a marketplace), dedicated mobile applications, and live GPS route optimization.
- **Future versions** may include native mobile applications, real-time GPS tracking of pickup vehicles, push notifications, and expanded machine-learning features such as demand forecasting for NGOs.

17. Risk Analysis

Risk	Impact	Mitigation
Incorrect or incomplete donation data (e.g., wrong expiry time)	Medium	Mandatory field validation and confirmation prompts before submission.
Database failure or data loss	High	Scheduled automated backups and transaction-safe JDBC operations.
Unauthorized access to donor/NGO accounts	High	Secure password hashing, role-based access control, and session management.
Low initial user adoption by donors or NGOs	Medium	Awareness campaigns, simple onboarding, and partnerships with local NGOs during launch.
Insufficient historical data for reliable ML matching at launch	Medium	Use a rule-based/decision-tree fallback ranking until enough pickup data is collected to train the Random Forest model.
Food safety concerns with donated items	High	Condition/expiry fields, donor verification, and a rating system to flag unreliable donors.

18. Conclusion

The Food Waste Donation Management System provides a practical and scalable solution for reducing food waste and improving food distribution in Bangladesh, a country where roughly a third of available food is lost or wasted each year despite widespread food insecurity. By digitizing and streamlining the donation process — from listing and intelligent NGO matching to pickup confirmation and impact tracking — the platform closes the gap between food surplus and unmet need using a fast, transparent, and trustworthy channel. Built on Java, JDBC, and MySQL, and enhanced with a Random Forest-based recommendation engine for NGO matching, the system offers secure, maintainable donation management while remaining well-positioned for future growth, including mobile applications, GPS-based logistics, and deeper AI-driven impact forecasting.